Offline Personal Assistant — Project Guide

Architecture, RAG workflow, privacy, and prototype milestones

1. Project summary

The Offline Personal Assistant is an offline-first assistant for personal information and everyday tasks. It can save and retrieve local memories, manage reminders, answer questions about local documents, and use online or connected services for information that requires current data.

2. Capability map

Capability	What it does	Network needed?
Local memory	Stores and retrieves user-approved facts or preferences on the device.	No, if local storage is available
RAG document search	Finds relevant passages in PDFs, notes, and text files, then answers with source citations.	No, if models and index are local
Reminders	Stores reminders and due dates locally and notifies the user using supported device mechanisms.	Usually no
Weather / current facts	Retrieves fresh information from online sources.	Yes
Outlook summaries	Summarizes emails the user has authorized the assistant to access.	Usually yes
GitHub summaries	Summarizes repositories, issues, pull requests, or activity from connected accounts.	Usually yes

3. Proposed RAG architecture

Document import → text extraction → chunking → local embeddings → vector index → query embedding → relevant chunk retrieval → local language model → answer with citations.

4. Suggested prototype stack

• Document parsing: PyMuPDF for PDF files, python-docx for Word files, and standard text readers.
• Embeddings: a compact local sentence-transformers model such as all-MiniLM-L6-v2, if suitable for the target device and language.
• Vector search: FAISS or Chroma running locally.
• Metadata: SQLite or a structured local store.
• Answer generation: the team's existing local LLM or a locally available model served through a compatible runtime.

5. Privacy and reliability principles

• Keep private documents and indexes local by default.
• Preserve filename, section, page number, and chunk ID as metadata.
• Show citations for answers based on retrieved material.
• Say when evidence is missing; do not invent dates, decisions, or names.
• Treat retrieved document content as untrusted data, not as system instructions.
• Check permissions before indexing or retrieving connected emails and repository content.
• Provide a way to delete a source and its associated chunks and embeddings.

6. Implementation milestones

Milestone	Deliverable	Done when
M1	Import and parse one Markdown/PDF file	Extracted text is visible
M2	Chunk and embed text	Chunks and metadata are stored
M3	Retrieve top relevant passages	Test queries find expected passages
M4	Generate grounded answers	Answers use retrieved context
M5	Citations and abstention	Sources display; unknown facts are not fabricated
M6	Offline test	Local document Q&A; works with network disabled

7. Demo acceptance tests

- Ask a question whose answer is explicitly in the document and check its citation.
- Ask a question that requires two sections or documents.
- Ask for a deadline that is not specified and verify that the assistant says it is not found.
- Disconnect the network and repeat a local-document question.
- Change a source file and verify that re-indexing updates the answer.

This is a proposed guide for the hackathon prototype based on the project description supplied for this demo. Replace assumptions with the team's actual implementation details.